Area in Polar Coordinates

ANSWER KEY

Section 1: Sector Area Formula & Single Curves

1 Which integral formula calculates the area of a region bounded by the polar curve $r = f(\theta)$ from α to β?

a. $\frac{1}{2}\int_{\alpha}^{\beta} r^2 d\theta$ ✓

b. $\int_{\alpha}^{\beta} r^2 d\theta$

c. $\frac{1}{2}\int_{\alpha}^{\beta} r d\theta$

d. $\pi\int_{\alpha}^{\beta} r^2 d\theta$

Polar area is the sum of infinitesimal circular sectors with area $dA = \frac{1}{2}r^2 d\theta$.

2 Find the area enclosed by the circle $r = 4\sin\theta$ for $\theta \in [0, \pi]$.

a. 4π ✓

b. 8π

c. 16π

d. 2π

Area $= \frac{1}{2}\int_{0}^{\pi} 16\sin^2\theta d\theta = 8[\frac{\theta}{2} - \frac{\sin(2\theta)}{4}]_0^{\pi} = 8(\frac{\pi}{2}) = 4\pi$.

3 What is the area enclosed by one petal of the four-petal rose $r = 2\cos(2\theta)$?

a. $\frac{\pi}{2}$ ✓

b. π

c. 2π

d. $\frac{\pi}{4}$

One petal spans from $\theta = -\pi/4$ to $\pi/4$. Area $= \frac{1}{2}\int_{-\pi/4}^{\pi/4} 4\cos^2(2\theta) d\theta = 2\int_{-\pi/4}^{\pi/4} \frac{1+\cos(4\theta)}{2} d\theta = [\theta + \frac{\sin(4\theta)}{4}]_{-\pi/4}^{\pi/4} = \frac{\pi}{4} - (-\frac{\pi}{4}) = \frac{\pi}{2}$.

Section 2: Area Between Two Polar Curves

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4 Which integral represents the area inside $r = R(\theta)$ and outside $r = r(\theta)$?

a. $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta)^2 - r(\theta)^2)d\theta$ ✓

b. $\frac{1}{2}\int_{\alpha}^{\beta}(R(\theta) - r(\theta))^2 d\theta$

c. $\int_{\alpha}^{\beta}(R^2 - r^2)d\theta$

d. $\frac{1}{2}\int_{\alpha}^{\beta}R(\theta)r(\theta)d\theta$

Area is difference of sectors: $\frac{1}{2}\int(R^2 - r^2)d\theta$.

5 Find the points of intersection of $r = 3$ and $r = 6\cos\theta$.

a. $\theta = \pm\frac{\pi}{3}$ ✓

b. $\theta = \pm\frac{\pi}{6}$

c. $\theta = \pm\frac{\pi}{4}$

d. $\theta = \frac{\pi}{2}$

$6\cos\theta = 3 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \pm\frac{\pi}{3}$.

6 Set up the integral for the area inside $r = 6\cos\theta$ and outside $r = 3$.

a. $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(36\cos^2\theta - 9)d\theta$ ✓

b. $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(6\cos\theta - 3)^2 d\theta$

c. $\int_{0}^{\pi/3}(36\cos^2\theta - 9)d\theta$

d. $\frac{1}{2}\int_{0}^{\pi/3}(6\cos\theta - 3)d\theta$

Outer curve is $6\cos\theta$, inner curve is 3. Formula: $\frac{1}{2}\int_{-\pi/3}^{\pi/3}(36\cos^2\theta - 9)d\theta$.

Section 3: Common Interior of Two Curves

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7 How is the common interior area of two overlapping polar curves calculated when the boundary changes?

a. **By finding the intersection angle and splitting the region into separate single-curve integrals** ✓

b. By subtracting the outer curve from the inner curve

c. By integrating the product of the two curves

d. By averaging the two individual areas

When rays from the origin transition from hitting one curve to hitting another, the area must be split at the intersection angle into separate integrals.

8 Find the area inside both $r = 2$ and $r = 4\sin\theta$ in the first quadrant.

a. $\frac{4\pi}{3} - \sqrt{3}$ ✓

b. 2π

c. $\frac{2\pi}{3}$

d. $4\sqrt{3}$

Intersection: $4\sin\theta = 2 \Rightarrow \sin\theta = 1/2 \Rightarrow \theta = \pi/6$. From 0 to $\pi/6$, bound is $4\sin\theta$; from $\pi/6$ to $\pi/2$, bound is 2. Sum yields $\frac{4\pi}{3} - \sqrt{3}$.

9 Why does the pole require special care when finding intersection points of polar curves?

a. **Because $r = 0$ corresponds to the same physical point (the origin) regardless of the angle θ** ✓

b. Because θ is undefined at $r = 0$

c. Because the area at the pole is infinite

d. It does not require special care

The origin can be reached by two curves at entirely different values of θ, so setting $f(\theta) = g(\theta)$ may fail to detect an intersection at the pole.

Section 4: Integration Limits & Full-Period Curves

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10 What is the range of θ required to trace the limaçon $r = 2 + \cos\theta$ exactly once?

a. $[0, 2\pi]$ ✓

b. $[0, \pi]$

c. $[0, 4\pi]$

d. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

For a limaçon without loops ($r = a + b\cos\theta$ with $a > b$), the curve is traced exactly once as θ traverses $[0, 2\pi]$.

11 Find the total area enclosed by the cardioid $r = 2(1 + \cos\theta)$.

a. 6π ✓

b. 4π

c. 8π

d. 12π

Area $= \frac{1}{2}\int_0^{2\pi} 4(1 + 2\cos\theta + \cos^2\theta)d\theta = 2\int_0^{2\pi}(1 + 2\cos\theta + \frac{1+\cos(2\theta)}{2})d\theta = 2(2\pi + 0 + \pi) = 6\pi.$

12 How many times is the circle $r = \sin(2\theta)$ traced over $[0, 2\pi]$?

a. **It is a 4-petal rose traced once over** $[0, 2\pi]$ ✓

b. It is a circle traced twice

c. It is traced 4 times

d. It is an ellipse

$r = \sin(2\theta)$ is an eight-petal or four-petal curve (four distinct petals) traced once completely as θ increases from 0 to 2π.

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